DeltaCodes: AI-Powered Healthcare Policy Change Highlighter **Jaehoon Rhie**

Defining the Problem (Topic)

Healthcare payment systems depend on rules and codes that change regularly across multiple sources. For example, the American Medical Association (AMA) issues annual Current Procedural Terminology (CPT) updates. CPT 2026 alone introduced 418 total changes, including 288 new codes, 84 deletions, and 46 revisions effective January 1, 2026 (American Medical Association & American Medical Association, 2025).

Policy management on keeping up with the changes is not as simple as a single refresh that could be automated annually, but organizations are expected to keep up with continuously changing policies across multiple sources which may conflict and overlap with one another.

Human review still exists in this pipeline as not every change is straightforward and requires thorough analysis. Some changes also require interpretation such as exceptions applying in a specific context or some rules taking precedence when there's conflict. The current bottleneck sits immediately before the approval step, where a policy analyst must manually compare two versions of a document to identify what changed, assess ambiguity, and hand over structured instructions to engineers for system updates and automation. This process is slow and hard to scale.

Industry Trends

The healthcare technology industry is actively moving towards layered architectures and automation with AI technology. From a recent review by Maimaitiaili et al. (2025), 29 hospital implementations are analyzed into a 5-layer model (Infrastructure, Data, Algorithm, Application, Security). This paper highlights the importance of a systematic approach for scalability and to point out how the technical core (Data, Algorithm, Application) is successfully integrated but security and compliance parts are isolated.

An article about regulatory technology also addresses how healthcare organizations are under constant pressure to meet regulatory standards, minimize risks, and protect patient safety (TAdviser, n.d.).

Opportunities and Threats

The opportunity is reducing the manual comparison step in policy ingestion. An AI-assisted tool positioned at the intake stage of the pipeline can analyze what changed, classify changes by type and confidence, and flag ambiguous points before a human analyst takes a look at the source documents. This significantly reduces the manual review time, risk of missed changes, and produces a structured change log that engineering teams can act on directly. As the update cycles become more frequent, the productivity gains from this tool compound over time.

The primary threat is LLM hallucination. The risk that a model confidently identifies or describes a change that did not occur, or misclassifies the nature of a change in a way that leads to incorrect rule updates. In a healthcare payment context, a hallucinated change that reaches production can cause systematic claim errors across large populations before it is caught.

Strategic Option for Cotiviti

Cotiviti could explore options in developing an internal policy classification tool, provisionally to sit at the front of its policy ingestion pipeline. The tool would take two versions of a policy PDF. For example, the CMS Stark Law CPT/HCPCS code list for 2025 and 2026, extract the text, and use a large language model to identify, classify, and summarize what changed between versions. Each detected change would be assigned a change type, a confidence level, and an impact rating. Changes flagged as low confidence or ambiguous would be automatically routed to mandatory human review rather than passed downstream. The tool would output a structured report for the reviewing analyst.

Possible threats of hallucination risk are mitigated by analysts sitting in the loop as the tool does not directly make changes or decisions such as reimbursement decisions or auto approval.

The recommended next step is tracking meaningful metrics with success measured by reduction in analyst review time and reduction in changes that reach engineering without prior human classification. This positions Cotiviti to take the first step in building a fully layered policy ingestion architecture while managing the governance and compliance risks.

References

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